



CSE 323: Operating Systems Design

Section 4

Quiz 2

Total Points: 20 (4 Questions)

Allowed Time: 30 minutes

NAME:

ID:

Q1 [5 Points] Consider a multithreaded process (P_0) containing three threads (T_0, T_1, T_2) running on a single-core CPU.

- a. Can T_0 and T_2 run in parallel? Why or why not? [2 Points]

No. A single core can only process one Thread at a time
For true Parallelism, we need multiple cores

- b. Execution Times for the threads are as follows:

Thread	Execution Time (nano-seconds, ns)
T_0	3
T_1	1
T_2	4

Explain why the Total time to complete P_0 would be greater than 8ns [2 Points]

Time in between i.e Context Switch Overhead accumulates when switching between threads. Need to copy new values to registers.

- c. Which is faster: creating a new thread within the same process or executing the fork() function? [1 Point]

Creating a new thread.

Q2 [5 Points] Consider a Many-to-One relationship between User Threads and a Kernel Thread.

- a. Why do user threads need to map to the kernel thread? Is this the only possible mapping? [2 Points]

There must be a relation between user & kernel threads because user threads need kernel support to work e.g system calls.
No, there are other mappings possible like one to one & many to many

- b. What happens if one user thread is currently using the kernel thread and another user thread initiates a system call? [2 Point]

In this mapping, only one thread can initiate a system call at a time & the entire process blocks until that system call is resolved. The other user thread needs to wait.

c. What symbol do we use in diagrams to represent a thread? [1 Point]

} (Squiggly line)

Q3 [5 Points] The following set of processes arrive at various times on the Ready Queue to be issued on a single core CPU. The Ready Queue is implemented using the **First Come, First Served** algorithm. Draw the Gantt Chart and show the **Waiting Times** for each of the processes.

Process	Arrival Time	Burst Time	<u>CT</u>	<u>TAT</u>	<u>WT</u>
P ₁	00	5	5	5	0
P ₂	00	2	7	7	5
P ₃	01	3	10	9	6
P ₄	12	2	14	2	0
P ₅	02	1	11	9	8

0 5 7 10 11 12 14

Q4 [5 Points] The following set of processes arrive at various times on the Ready Queue to be issued on a single core CPU. The Ready Queue is implemented using the **Shortest Job First with Preemption** algorithm. Draw the Gantt Chart and show the **Response Times** for each of the processes.

Process	Arrival Time	Burst Time	<u>CT</u>	<u>TAT</u>	<u>WT</u>	<u>RT</u>
P ₁	1	4	12	11	7	7
P ₂	0	4	8	8	4	4
P ₃	1	1	2	1	0	0
P ₄	0	3	7	7	1	0

0 1 2 7 8 12